Intrusion Detection in IoT Networks Using Machine Learning and Deep Learning: Experiments on CIC IoT-DIAD 2024 Dataset

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***Abstract*—The exponential growth of Internet of Things (IoT) devices has increased the vulnerability of networks to sophisticated cyberattacks. Traditional security mechanisms struggle to cope with the scale and diversity of IoT traffic, motivating the adoption of Machine Learning (ML) and Deep Learning (DL)-based intrusion detection systems (IDS). This paper presents a comparative study of multiple ML and DL models for intrusion detection using a subset of the CIC IoT-DIAD 2024 dataset. We implemented classical ML algorithms (Random Forest, Logistic Regression, Decision Tree, Naïve Bayes, KNN, Gradient Boosting), boosting methods (LightGBM, XGBoost), and DL architectures (Fully Connected Deep Neural Network and 1D-CNN). To address class imbalance, Synthetic Minority Over-Sampling Technique (SMOTE) was applied. Experimental results demonstrate that ensemble approaches outperform traditional methods, achieving high accuracy and robust multi-class detection across 11 attack categories. This study highlights the effectiveness of hybrid approaches for securing IoT environments.**

***Index Terms*—IoT, Intrusion Detection Systems, Machine Learning, Deep Learning, Anomaly Detection, Cybersecurity.**

I. Introduction

The proliferation of Internet of Things (IoT) devices has transformed modern life by enabling smart homes, connected healthcare, and industrial automation. However, the heterogeneity and resource constraints of IoT devices make them attractive targets for cyberattacks such as denial of service, probing, and data exfiltration. Effective Intrusion Detection Systems (IDS) are crucial for safeguarding IoT networks.

Recent studies show that traditional rule-based IDS fail to generalize against zero-day attacks. Machine Learning (ML) and Deep Learning (DL) have emerged as promising solutions, leveraging data-driven approaches to detect complex attack patterns. However, the challenges of imbalanced datasets, real-time detection requirements, and high-dimensional IoT traffic remain unresolved.

This paper addresses these challenges by evaluating multiple ML and DL approaches on the CIC IoT-DIAD 2024 dataset. The contributions of this work are:

1. Implementation of diverse ML, boosting, and DL models for IoT intrusion detection.

2. Application of SMOTE to handle severe multi-class imbalance in attack categories.

3. Comparative evaluation of model performance using standard metrics (accuracy, precision, recall, F1-score).

4. Insights into the strengths of ensemble approaches in IoT security.

II. Literature Review

In recent years, the need for effective intrusion detection methods in IoT networks has grown, and various anomaly-based detection models have been proposed. These models are designed to detect previously unknown attacks by identifying abnormal behavior in the system rather than relying on predefined attack signatures.

Bhavsar et al. [1] present an anomaly-based intrusion detection system (IDS) tailored for IoT applications, leveraging a Pearson-Correlation Coefficient Convolutional Neural Network (PCC-CNN) architecture. The method integrates feature selection via Pearson correlation with a CNN classifier to detect anomalies in both binary and multiclass settings. Tested on NSL-KDD, CICIDS-2017, and IOTID20 datasets, the model achieves up to 99.89% accuracy with a minimal misclassification rate of 0.001, outperforming traditional PCC-based ML models such as KNN, CART, and SVM. The authors emphasize that conventional signature-based IDS approaches fail to detect novel attacks effectively due to their reliance on predefined patterns, making anomaly-based detection preferable for dynamic IoT environments. The PCC-CNN's lightweight nature makes it suitable for real-time deployment in resource-constrained IoT systems, addressing critical cybersecurity threats such as DDoS, spoofing, and MITM attacks. This research demonstrates that coupling statistical feature extraction with deep learning significantly enhances IDS performance in IoT contexts, providing improved detection rates and reduced false alarms compared to conventional ML approaches. Overall, the study advances IoT network defense by bridging the gap between accuracy, adaptability, and computational efficiency. .

Lazzarini et al. [2] propose an advanced ensemble architecture to improve IDS performance in heterogeneous IoT networks. The stacking method combines multiple deep learning base learners, such as Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and Deep Neural Networks , whose outputs are aggregated by a meta-learner for final prediction. This approach addresses the limitations of single deep models, particularly their susceptibility to overfitting and dataset bias. Using benchmark IoT security datasets, the ensemble demonstrates superior accuracy and robustness in detecting diverse cyberattacks, including DDoS, probing, and botnet intrusions. The model also exhibits reduced false positive rates, an essential factor for practical IoT deployment where frequent alerts can overwhelm administrators. The study highlights that ensemble methods exploit the complementary strengths of different deep architectures, leading to improved generalization across attack types. The authors recommend further optimization for real-time deployment, such as reducing computational load without compromising detection accuracy. This work contributes to the ongoing evolution of IDS design, emphasizing model diversity and meta-learning as key factors in enhancing IoT network security. .

Baksh et al. [3] proposes a deep learning–based intrusion detection system (IDS) specifically designed for securing Internet of Things (IoT) networks. The authors explore and evaluate three different deep neural network architectures, Feed-Forward Neural Network (FFNN), Long Short-Term Memory network (LSTM), and Random Neural Network (RandNN) to detect anomalous or malicious traffic in IoT environments. Each model leverages its unique strengths: the FFNN efficiently captures complex patterns in network data, the LSTM excels at modeling temporal dependencies, and the RandNN offers dynamic adaptability through its random connectivity and flexible behavior. Performance evaluation using the CIC-IoT22 dataset demonstrates that all three models outperform existing state-of-the-art deep learning IDS approaches. The FFNN model achieves the highest accuracy at approximately 99.93%, followed by the LSTM at 99.85%, and the RandNN at 96.42%. These findings underline the potential of deep learning to strengthen IoT security by accurately detecting various cyber threats in network traffic. By comparing and validating multiple architectures, the paper provides valuable insights for developing efficient, reliable, and adaptive IDS solutions for increasingly interconnected IoT systems. .

In their comprehensive study, Kok et al. [4] present a detailed examination of intrusion detection systems (IDS) through the lens of machine learning (ML) methodologies. The authors underscore the growing significance of ML in cybersecurity, particularly in enhancing the detection capabilities of IDS by improving accuracy and minimizing false alarms. The review highlights the dependence of model performance on the nature of the dataset and the chosen algorithm, emphasizing the strong influence of data structure on algorithm suitability. Despite the evolution of threats and technologies, the study notes a concerning reliance on outdated datasets such as KDDCup99 and NSL-KDD, which may hinder the relevance and robustness of IDS models in contemporary environments. The paper categorizes ML algorithms into 11 types, including decision trees, neural networks, and ensemble methods, and reports a trend towards hybrid and soft computing techniques due to their improved performance. Most studies reviewed focus on classification tasks using supervised learning, which may limit the ability to detect novel or modified attacks. Evaluation metrics such as accuracy, True Positive Rate (TPR), and False Positive Rate (FPR) are identified as most frequently used in assessing IDS models. Kok et al. recommend exploring newer datasets like CICIDS2017 and adopting clustering techniques to better detect anomalies, thereby enhancing the adaptability and effectiveness of IDS in real-world scenarios. .

The paper by Mahipal and Khan [5] offers a forward-looking analysis of how artificial intelligence (AI) is reshaping the cybersecurity landscape, both in terms of threat evolution and defense capabilities. The authors stress that while AI provides substantial improvements in detecting, predicting, and responding to cyber threats, it also introduces new vulnerabilities and attack vectors. Their review identifies a dual-edged impact of AI: it enables sophisticated automated defense systems, yet it also empowers adversaries to execute intelligent, adaptive, and hard-to-detect cyberattacks. The paper outlines how AI technologies, such as machine learning and deep learning, are being deployed to enhance threat detection systems, anomaly detection, and real-time response mechanisms. However, it also warns of the emerging risks posed by AI-driven malware and adversarial machine learning attacks that can bypass conventional security measures. A significant contribution of the study lies in its discussion of ethical challenges, such as data privacy and algorithmic bias, which must be addressed to ensure responsible AI adoption in cybersecurity. The authors call for a balanced and collaborative approach to policy, research, and implementation, stressing the need for adaptive AI-based security frameworks that can evolve in response to dynamic threat environments. .

In their study, Kumar and Rao [6] propose a comprehensive cybersecurity attack detection model grounded in machine learning (ML) techniques. The paper emphasizes the critical need for automated, intelligent systems to identify and mitigate cyber threats in real-time, given the growing volume and complexity of modern attacks. The authors begin by classifying cyberattacks into five main categories probing, denial-of-service (DoS), user-to-root (U2R), remote-to-local (R2L), and normal traffic highlighting the importance of accurately identifying these patterns to maintain secure digital environments. The model utilizes both supervised and unsupervised machine learning algorithms, such as Decision Trees, Random Forests, and Support Vector Machines (SVM), and evaluates their performance based on accuracy, precision, recall, and F1-score. The experimental analysis uses the NSL-KDD dataset, a widely accepted benchmark in intrusion detection research. The study reports that Random Forest outperforms other models in terms of detection accuracy and resilience to overfitting,

suggesting its robustness for real-world deployment. Furthermore, the authors stress the importance of feature selection and data preprocessing, noting that effective detection depends heavily on high-quality, well-structured input data. The paper concludes by advocating for hybrid and ensemble methods as future directions to enhance detection performance and reduce false positives. .

III. Methodology

A. *Dataset Description*

The dataset used in this study was the IoT-DIAD 2024 dataset [7] available from the Canadian Institute for Cybersecurity (CIC) at the University of New Brunswick. The dataset includes a comprehensive collection of IoT network traffic data and various types of attacks that are commonly observed in IoT networks, such as Distributed Denial-of-Service (DDoS), spoofing, and Man-in-the-Middle (MITM) attacks. This dataset is specifically designed to aid in the development and evaluation of intrusion detection systems for IoT environments. The data is structured to allow for both supervised and unsupervised learning approaches in IDS model training and testing.

There are a lot of attack categories in this dataset, so we took 1 benign class and 10 attack classes and merged their csv files to create a dataset csv file. We could not take more data available in the dataset due to computational limitations. There were 84 features in the dataset.

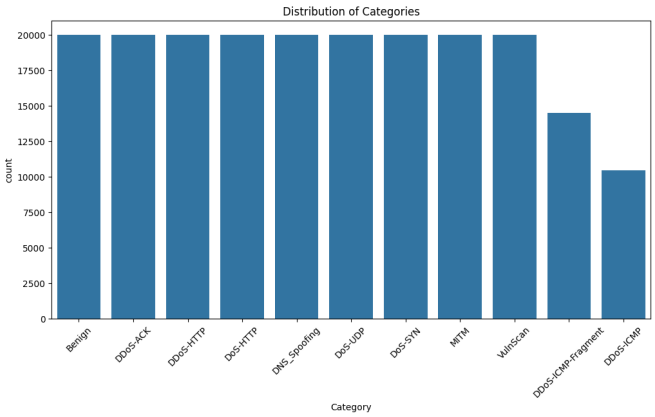

Fig. 1. Sample number of the classes in the merged dataset.

B. *Data Preprocessing*

Prior to model training, the dataset was carefully preprocessed to improve learning quality and mitigate noise. Instances containing missing values, duplicates, or infinite entries were removed to ensure data integrity. A correlation matrix was computed, and 27 highly correlated features were eliminated to reduce redundancy and prevent multicollinearity. Irrelevant identifiers such as Flow ID, Source IP, Destination IP, Timestamp, and Category were also discarded, as they were irrelevant. To address differences in feature scales, StandardScaler was applied, transforming features to zero mean and unit variance, which benefits both distance-based algorithms and gradient-based optimization. Furthermore, the dataset's class imbalance was addressed using SMOTE, which generates synthetic samples for minority classes, thereby improving model generalization and recall. Finally, a stratified 70/30 train-test split was applied to preserve the relative class distribution across both sets.

C. *Model Development*

We implemented a diverse set of Machine Learning (ML) and Deep Learning (DL) models to evaluate their effectiveness in IoT intrusion detection. The ML models included Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), k-Nearest Neighbors (KNN), and Naïve Bayes (NB), which serve as classical baselines. In addition, we employed Gradient Boosting (GB), XGBoost, and LightGBM, which are ensemble learning methods known for handling high-dimensional data and improving generalization.

For Deep Learning, two architectures were implemented: a Fully Connected Deep Neural Network (DNN) and a 1D Convolutional Neural Network (1D-CNN). The DNN consisted of multiple dense layers with ReLU activations, dropout regularization to mitigate overfitting, and a softmax output layer for multi-class classification. The 1D-CNN was designed to capture local sequential patterns in network traffic features, employing convolutional and pooling layers followed by fully connected layers. Both models were trained using the Adam optimizer with categorical cross-entropy loss and early stopping criteria to prevent overfitting.

All models were trained on the preprocessed dataset with a 70/30 train-test split. Hyperparameters for the ML algorithms (e.g., number of estimators in RF, learning rate in boosting models, k in KNN) were optimized empirically using grid search and cross-validation. For DL models, batch size and epoch limits were tuned experimentally to balance training efficiency and performance. This diverse set of models enables a comprehensive comparison of traditional ML, ensemble methods, and deep learning approaches in the context of IoT intrusion detection.

D. *Evaluation Metrics*

To comprehensively assess model performance, both overall accuracy and class-specific metrics were employed. Accuracy provides a general measure of correct classifications; however, due to class imbalance, it alone is insufficient. Therefore, Precision, Recall, and F1-score were calculated for each class and averaged using both macro (treating all classes equally) and weighted (accounting for class frequencies) strategies. Precision measures the proportion of correctly predicted positive samples, Recall evaluates the ability to detect actual positive samples, and F1-score provides a harmonic balance between the two. In addition, confusion matrices were generated to visualize per-class classification strengths and weaknesses. For deep learning models, training and validation accuracy/loss curves were also monitored to detect underfitting or overfitting.

Together, these metrics provide a robust evaluation framework for multi-class intrusion detection.

E. Results and Analysis

TABLE I
COMPARISON OF ML AND DL MODELS ON IOT-DIAD 2024 DATASET

Model	Accuracy (%)	Precision	Recall	F1-score
Logistic Regression	71.0	0.71	0.71	0.70
Random Forest	92.0	0.92	0.92	0.92
KNN	87.0	0.87	0.87	0.87
Decision Tree	90.0	0.90	0.90	0.90
Naive Bayes	50.0	0.50	0.50	0.47
Gradient Boosting	85.0	0.85	0.85	0.85
LightGBM	93.0	0.93	0.92	0.92
XGBoost	92.0	0.92	0.92	0.92
DNN	80.0	0.81	0.80	0.79
1D CNN	80.0	0.81	0.80	0.80

The comparative performance of ML and DL models on the IoT-DIAD 2024 dataset is summarized in Table I. Traditional classifiers such as Naïve Bayes and Logistic Regression performed poorly, with accuracies of 50% and 71%, respectively, highlighting their limited capability in handling the high-dimensional and imbalanced nature of IoT traffic. KNN, Decision Tree, and Gradient Boosting achieved moderate results, with accuracies between 85–90%, showing improved detection capability but still struggling in certain attack classes. In contrast, ensemble approaches such as Random Forest, LightGBM, and XGBoost consistently outperformed traditional methods, achieving accuracies above 92% and balanced precision, recall, and F1-scores. These results emphasize the advantage of ensemble models in capturing complex feature interactions and improving generalization.

Fig. 2. Training and validation accuracy/loss curves for the DNN model.

For Deep Learning approaches, both the DNN and 1D-CNN achieved competitive but slightly lower performance (80%) compared to ensemble methods. This can be attributed to computational constraints that limited deeper architectures and extensive hyperparameter tuning. However, DL models demonstrated the potential to capture latent feature representations, which could be further leveraged in future work with optimized architectures. Overall, the findings highlight that ensemble ML models, particularly LightGBM, provide the most robust and reliable solution for multi-class IoT intrusion detection in the evaluated setting.

Fig. 3. Training and validation accuracy/loss curves for the 1D-CNN model.

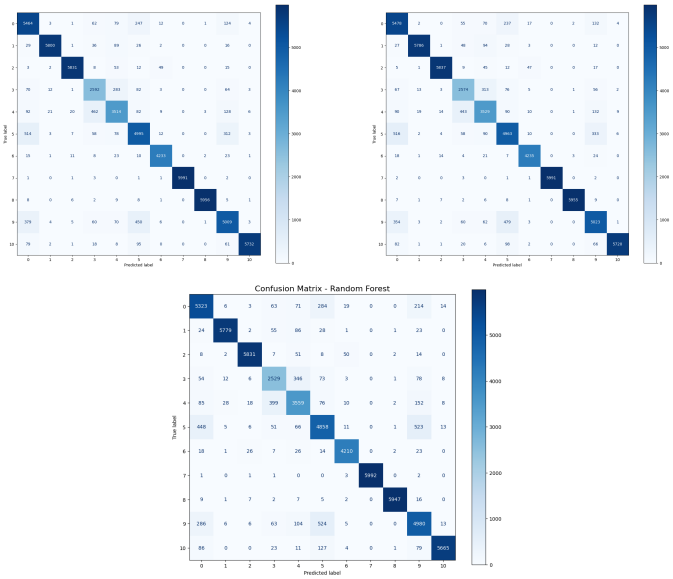

Fig. 4. Confusion matrices for LightGBM (top left), XGBoost (top right), and Random Forest (bottom) on the IoT-DIAD 2024 dataset.

The confusion matrices provide further insight into class-level performance. LightGBM and XGBoost achieved consistently high true positive rates across most attack categories, with minor misclassifications between similar traffic patterns, such as DDoS and probing. Random Forest showed slightly higher confusion in distinguishing between benign traffic and certain low-frequency attacks, likely due to residual class imbalance. Overall, the matrices confirm that ensemble approaches achieve both high overall accuracy and balanced detection across diverse attack categories.

IV. CONCLUSION

The experimental results demonstrate that ensemble models, particularly LightGBM, provide the most reliable performance for multi-class IoT intrusion detection, achieving high accuracy, precision, recall, and F1-scores across all evaluated attack categories. Boosting methods effectively capture complex

feature interactions and mitigate class imbalance, which is prevalent in IoT traffic. Traditional classifiers, such as Naïve Bayes and Logistic Regression, showed limited capability, while KNN, Decision Tree, and Gradient Boosting achieved moderate results.

Deep learning models, including the DNN and 1D-CNN, demonstrated competitive performance and the ability to extract latent feature representations, although their results were slightly lower due to computational constraints and the limited dataset size. Analysis of confusion matrices and training/validation curves further confirmed that ensemble models maintain robust and balanced detection across diverse attack categories, whereas deep learning models occasionally misclassify similar attack patterns.

Despite these promising results, this study has some limitations. Only a subset of the CIC IoT-DIAD 2024 dataset was used, and real-time or resource-constrained deployment scenarios were not evaluated. Future work could focus on larger datasets, hybrid ensemble–deep learning architectures, real-time streaming detection, model compression for IoT deployment, and explainable AI methods to improve interpretability and trust in automated intrusion detection systems.

Overall, the findings highlight the superiority of ensemble approaches for IoT intrusion detection while demonstrating the potential of deep learning models for capturing complex traffic patterns. These insights provide a foundation for developing more efficient, robust, and interpretable IDS solutions for increasingly connected IoT environments.

REFERENCES

[1] M. Bhavsar, K. Roy, J. Kelly, and O. Odeyomi, "Anomaly-based intrusion detection system for IoT application," *Discover Internet of Things*, vol. 3, 2023, doi: 10.1007/s43926-023-00034-5.

[2] R. Lazzarini, H. Tianfield, and V. Charissis, "A stacking ensemble of deep learning models for IoT intrusion detection," *Knowledge-Based Systems*, vol. 279, pp. 110941, 2023, doi: 10.1016/j.knosys.2023.110941.

[3] S. Allah Bakhsh, M. A. Khan, F. Ahmed, M. Alshehri, H. Ali, and J. Ahmad, "Enhancing IoT network security through deep learning-powered Intrusion Detection System," 2023.

[4] S. Kok, A. Azween, N. Jhanjhi, and M. Supramaniam, "A review of intrusion detection system using machine learning approach," *International Journal of Engineering Research and Technology*, vol. 12, pp. 8-15, 2019.

[5] R. Ratnawita, "Cybersecurity in the AI Era Measures Deepfake Threats and Artificial Intelligence-Based Attacks," *Journal of the American Institute*, vol. 2, pp. 180-189, 2025, doi: 10.71364/s3emxx77.

[6] İ. Avci and M. Koca, "Cybersecurity Attack Detection Model, Using Machine Learning Techniques," *Acta Polytechnica Hungarica*, vol. 20, pp. 29-44, 2023, doi: 10.12700/APH.20.7.2023.7.2.

[7] M. Rabbani, J. Gui, F. Nejati, Z. Zhou, A. Kaniyamattam, M. Mirani, G. Piya, I. Opushnyev, R. Lu, and A. A. Ghorbani, "Device Identification and Anomaly Detection in IoT Environments," *IEEE Internet of Things Journal*, Dec